Table 2. Murine versus Human Salivary Gland Organoids

Feature	Murine organoids	Human organoids
Acinar differentiation	Acinar-like differentiation achievable under selected conditions; extent varies by gland and protocol.	Often ductal/basal-enriched during expansion; acinar differentiation remains protocol- and niche-dependent.
Typical morphology	Lumenized, cystic or branched structures; acinar-like clusters in some protocols.	Cystic, solid/dense or branched structures, depending on source and culture conditions.
Lineages / markers	Acinar, ductal and myoepithelial features reported; AQP5, BHLHA15/MIST1, AMY, mucins.	KRT5/KRT14 basal and KRT7/KRT19 ductal features; AQP5, BHLHA15 and AMY induction varies.
Secretory function	Secretagogue-responsive calcium signals, swelling and amylase activity reported in selected models.	Calcium and swelling responses demonstrated; mature secretory output remains variable and incompletely benchmarked.
ECM / media requirements	Outcomes depend on gland source, medium, passage and ECM conditions, including batch effects.	Sensitive to donor/cell source, medium, ECM and the timing of differentiation cues.
Stem/progenitor plasticity	Lineage contributions and plasticity reported in defined developmental and injury settings.	Progenitor-like states and differentiation reported; matched cross-species comparisons remain limited.
Donor / model variability	Inbred strains can reduce variation; strain, age, gland and protocol still influence outcomes.	Donor age, gland source, pathology, prior injury and culture history contribute to heterogeneity.
Representative applications	Development, morphogenesis, lineage regulation, injury–repair and secretory physiology.	Patient-specific disease and cancer models, drug testing and regenerative translation.
Refs	13,14,61,77,96	13,21,23,77,79,89

Species differences are multifactorial, involving gland architecture, niche requirements, matrix responses, donor age, and pathology, rather than a single intrinsic defect of human stem cells. 13,23,77,93–95,99